

Compare Linear and Polynomial Regression using Python

Aim

To implement and compare **Linear Regression** and **Polynomial Regression** using Python and evaluate their performance.

Apparatus Required

- Google Colab / Jupyter Notebook
- Python 3
- Libraries:
 - NumPy
 - Matplotlib
 - Scikit-learn (sklearn)

Theory

Linear Regression fits a straight line to the data and is suitable when the relationship between the independent and dependent variables is linear.

Polynomial Regression extends Linear Regression by adding polynomial features, allowing it to model non-linear relationships.

Comparison:

- Linear Regression fits a straight line.
- Polynomial Regression fits a curve.
- Polynomial Regression generally provides better accuracy for non-linear data.

Algorithm

1. Import the required libraries.
2. Create a sample dataset.
3. Train the Linear Regression model.
4. Convert the features into polynomial features.

5. Train the Polynomial Regression model.
6. Plot both regression models.
7. Compare the results.

Code (Google Colab)

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures

# Sample dataset
X = np.array([[1],[2],[3],[4],[5],[6],[7],[8],[9],[10]])
y = np.array([1.5,3.8,6.2,8.5,11,14.5,18.2,22.8,28.1,34])

# Linear Regression
lin_reg = LinearRegression()
lin_reg.fit(X, y)

# Polynomial Regression (Degree = 2)
poly = PolynomialFeatures(degree=2)
X_poly = poly.fit_transform(X)

poly_reg = LinearRegression()
poly_reg.fit(X_poly, y)

# Plot
plt.scatter(X, y, color='red', label='Data')
```

```

plt.plot(X, lin_reg.predict(X), color='blue', label='Linear Regression')
plt.plot(X, poly_reg.predict(X_poly), color='green', label='Polynomial Regression')
plt.xlabel("X")
plt.ylabel("Y")
plt.title("Linear vs Polynomial Regression")
plt.legend()
plt.show()

```

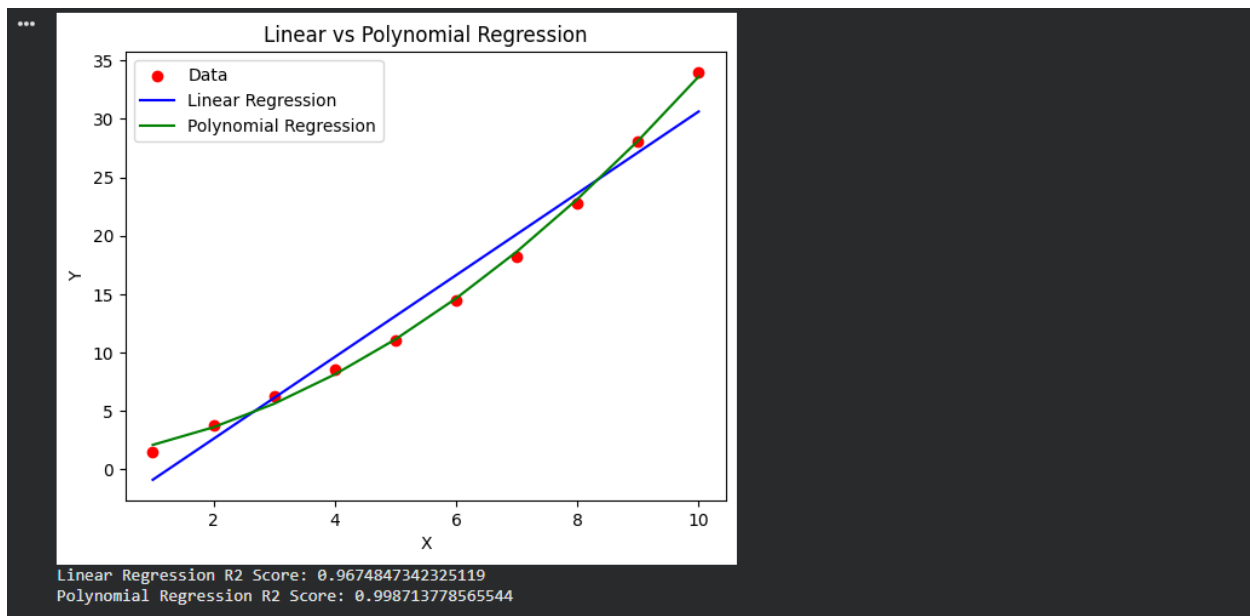
Scores

```

print("Linear Regression R2 Score:", lin_reg.score(X, y))
print("Polynomial Regression R2 Score:", poly_reg.score(X_poly, y))

```

Sample Output



Result

The **Linear Regression** and **Polynomial Regression** models were successfully implemented in Python. The **Polynomial Regression** model achieved a higher R^2

score and provided a better fit for the non-linear dataset compared to the Linear Regression model.